

Five Days at Fremont: What Does a Tesla Really Cost?

AC101 Introduction to Accounting – Part 1 integrated case (Sessions 1–5) *Worked one part per lecture through the week*

Before you start: what is real here, and what is not

This case is set inside a real company; it consists of both fact and fiction.

- **Tesla, Inc. is real**, and some of the financial data below comes from actual SEC filings and Tesla investor-relations reports; the period is stated on each exhibit. **All product costing scenarios, internal cost data, standards and strategic decisions in this case are entirely hypothetical.** They do not represent actual Tesla plans, costs or decisions, and any resemblance beyond the public filings is coincidental. The CFO, the colleagues and all internal dialogue are fictional; no real Tesla executive is depicted.
- **Every exhibit is labelled – REAL** (SEC 10-K FY2024, Compustat 2023–24, or Tesla IR, with the period stated), **SIMULATED** (with a one-line note on how the figures were constructed – industry benchmarks, teardown reports, the BloombergNEF battery index, or standards built on the real Q4 2024 base), or **ESTIMATE** (a statistic we computed from the real data, such as Part B's regression – not a figure Tesla discloses).
- **One stylised liberty, flagged where it is used (Part C):** all five models are placed at the Fremont factory for a clean single-factory example. In reality the Cybertruck is built at Giga Texas. Part C's margins are stylised levels – roughly 2.4 times Tesla's real 17.9% FY2024 gross margin – so read the *distribution* across models, not the level.
- **Battery figures in Part D are cell-level variable costs.** Do not compare them with pack-level prices (~\$115–140/kWh, BloombergNEF 2024), which add pack assembly, housing and battery management on top of the cell – a different scope.
- **Proxies are labelled.** "Revenue per car delivered" spreads energy, services and regulatory-credit revenue over cars delivered – it is *not* an average selling price and sits above the ~\$43k Model Y sticker. Part B's regression is estimated on Compustat's COGS series; Compustat reports COGS with depreciation, amortisation and impairment shown separately (\$5,368m in FY2024), so the intercept understates Tesla's total fixed production cost. Part C's discontinuation reload is a labelled worst case.

How to use this case. One part per day: each lecture ends with a case block – read that day's part, work its questions in your groups, and we discuss the solutions together. Every question is discussed in the lecture. A ► marks a more advanced question. A calculator is all you need; where figures are in \$m, the question says so.

One note on the exam. Variance analysis (Part E) is taught in Part 1 but **examined at the final exam only** – it does not appear on the mid-session paper. Everything else here is mid-session material. Any sales-volume variance in this case is **measured on a contribution basis**, and says so.

Monday, 7:40 a.m.

The Fremont factory is louder than the recruitment video suggested. You are a newly hired analyst on Tesla's corporate finance team, on site for your first week, and you have been inside the plant for forty minutes – long enough to watch a robot lift a painted body shell onto a battery pack in one unhurried motion – when your phone buzzes.

It is the CFO. Call her Reyes; like every executive in this case, she is invented.

*"Welcome aboard. Simple question for your first week: are we making money on the Model Y?
Briefing Friday, 4 p.m. Show your working."*

It reads like a question with a single number for an answer. By Friday afternoon you will know that it is five different questions wearing one sentence – and that each of them needs a different "cost".

Tesla in 2024: the company behind the question

Tesla designs, builds and sells electric vehicles and energy products, selling directly to customers rather than through franchised dealers. The Fremont, California plant — a former GM–Toyota joint-venture factory that Tesla bought in 2010 — builds the Model S, Model 3, Model X and Model Y; the Cybertruck is built at Giga Texas.

The financial year you have walked into was bruising. In fiscal 2024 Tesla reported total revenues of **\$97.69bn** and delivered **1,789,226** vehicles (producing 1,773,443) — but the first quarter of 2024 brought the company's first year-on-year delivery decline since the pandemic quarter of Q2 2020, and a price war that began with a single \$13,000 cut to the Model Y in January 2023 was still grinding on. Growth was no longer doing the explaining. Cost had to.

That is the world your Monday message landed in.

Part A — Monday: cost of what?

Your first instinct is the obvious one: look it up. Tesla files audited accounts with the SEC; surely the cost of a Model Y is in there somewhere.

It is not — and discovering *why* it is not is the real work of Day 1.

The FY2024 income statement (Exhibit A-1) is strikingly compact. Revenue of \$97,690m; cost of revenues of \$80,240m; gross profit of \$17,450m; then \$5,150m of selling, general and administrative expense, \$4,540m of research and development, \$684m of restructuring, and an operating income of \$7,076m. Six numbers below the revenue line, for a company that builds cars, batteries, software and Superchargers on three continents. The accounts are built for *outsiders* — investors, lenders, regulators — who need standardised, comparable, backward-looking totals. They are financial accounting. Reyes's question is an *insider's* question — forward-looking, decision-shaped, and about one product — and answering it is management accounting, which the published accounts were never designed to do.

So you start where a management accountant starts: **cost of what?** A cost only means something once you fix the **cost object** — a product (one Model Y), a plant (Fremont), a customer (a rental fleet), a project (Cybertruck development). The same salary is "direct" to the factory but "indirect" to a single car. Object first, cost second.

Then you sort costs two ways, and each way exists because it answers a different manager's question:

- **By behaviour** — "**how does profit move with volume?**" **Variable costs** change in total with output (battery cells: more cars, more cells). **Fixed costs** stay flat in total across a sensible range of volumes (factory insurance costs the same at 400,000 or 500,000 cars).
- **By traceability** — "**what does this product cost?**" **Direct costs** can be traced economically to the cost object (the battery pack in *this* Model Y). **Indirect costs — overhead** — are shared and must be *allocated* by some rule (the plant manager's salary covers every model). How to allocate is Wednesday's problem, and it is a big one.

A colleague on the costing team, hearing it is your first day, sends you a welcome present: a list of ten Fremont costs "to see whether the new analyst can sort the post" (Exhibit A-2). Every cost sits on *both* dimensions at once — paint is variable *and* direct; the property tax is fixed *and* indirect — and the 2×2 the ten items build is the foundation of the whole week.

One more distinction matters before you can read Tesla's statements properly: **cost versus expense**. A cost is a resource given up to get something; it becomes an expense only when it is *used up* to earn revenue. Some costs expense immediately (today's advertising). Some expense gradually (a machine, through depreciation). And a car's build cost does something stranger: it waits on the balance sheet as **inventory** — an asset — and hits profit as **cost of goods sold** only when the car is sold. A Model Y built in December and sold in January carries its cost across the year-end as an asset. Nothing is lost; it is just not in *this* year's profit. Hold onto that lane of the cost journey — it comes back with teeth on Wednesday.

Late in the afternoon, a memo lands from another analyst (Exhibit A-3, illustrative round numbers – explicitly not Tesla’s actuals). At 300,000 cars, the accountants report a full unit cost of \$30,000 per car: \$20,000 variable plus \$10,000 of fixed cost per car (\$3.0bn spread over 300,000). The memo scales that \$30,000 to a 500,000-car plan. It looks diligent. It is wrong by two billion dollars – because “fixed cost per car” is an artificial number that slides with whatever volume you divide by (\$10,000 at 300,000 cars; \$6,000 at 500,000). Total fixed cost does not move; only the arithmetic does. For decisions, think in totals.

By Monday evening you can give Reyes the honest half-answer. **Tesla as a whole made money in 2024: \$17.45bn gross, \$7.08bn operating** – straight off Exhibit A-1. But “the cost of a Model Y” is not one number. The full average cost (with allocated factory and R&D), the incremental cost of one more car, and the avoidable cost of not making it are *different numbers answering different decisions* – and the published accounts reveal none of them. Recovering them is the rest of your week.

Assignment questions – Part A

1. From Exhibit A-1, express cost of revenues and gross profit as a percentage of revenue. If you were the CFO, which of the three big cost lines – cost of revenues, SG&A, R&D – would you attack first, and why?
2. Cost object: **one Model Y**. Classify each of Exhibit A-2’s ten costs on both dimensions – fixed/variable and direct/indirect – with a one-line reason for each. Check yourself against the classification table from Session 1.
3. A Model Y is built in December 2024 but sold in January 2025. At the 31 December year-end, is its build cost a cost or an expense – and where does it sit? When does it hit profit, and under what name?
4. Using Exhibit A-3: a colleague budgets for 500,000 cars by scaling the \$30,000 full unit cost. Compute their budget and the correct one, and explain in two sentences where the phantom \$2.0bn comes from.
5. ► The 10-K alone cannot say whether the Model Y is profitable. Name three different “costs of a Model Y” and match each to a decision it serves, one line each. (This is the map of Parts C and D.)
6. Exhibit A-4 gives one quarter’s cost ledger for the Model Y line – and it contains more than you need. (a) Cost object: **one Model Y**. Using only the rows that belong, compute the variable cost of building one Model Y (its variable manufacturing cost). List every row you leave out with a one-line reason for rejecting it, and state one assumption your figure relies on. (b) ► One of the rows you rejected in (a) is a genuine cost of building Model Ys, yet it stayed out of your figure. Name that row, give the one word that justifies leaving it out here, and name the different “cost of a Model Y” it would belong to. No calculation needed.

Exhibits – Part A

Exhibit A-1 – Tesla FY2024 income statement REAL – SEC 10-K, fiscal year 2024 ([SEC EDGAR filing index](#)).

Line item	\$m
Total revenues	97,690
Cost of revenues	(80,240)
Gross profit	17,450
Selling, general and administrative	(5,150)
Research and development	(4,540)
Restructuring and other	(684)
Operating income	7,076

Exhibit A-2 – Ten Fremont costs to classify SIMULATED – illustrative cost descriptions only (no figures); a hypothetical exercise, not Tesla’s cost records.

1. Paint and coatings
2. Fremont property tax
3. Tyres
4. Welding gas and machine lubricants (used across all models)

5. Depreciation on Model-Y-only welding robots
6. Plant manager's salary
7. Shipping to the customer
8. Quality-inspector salary
9. Aluminium for body panels
10. Factory canteen

Exhibit A-3 – The analyst's shortcut SIMULATED – illustrative round numbers chosen to make the mechanics obvious; explicitly not Tesla's actuals.

Item	Figure
Total fixed cost per period	\$3.0bn
Variable cost per car	\$20,000
Fixed cost per car at 300,000 cars	\$10,000
"Full unit cost" at 300,000 cars	\$30,000
Fixed cost per car at 500,000 cars	\$6,000
The memo's budget for 500,000 cars	500,000 × \$30,000

Exhibit A-4 – One quarter on the Model Y line SIMULATED – a hypothetical quarter's cost ledger for the Model Y line only, built from teardown reports and industry benchmarks; not Tesla's cost records. Figures are quarterly totals in \$m unless a per-car figure is stated; the battery figure is pack-level (about 75 kWh at benchmark pack cost, consistent with the ~\$115–140/kWh pack prices flagged on the first page).

Cost line	\$m
Battery cells and pack	1,536
Other direct materials (body, drive units, electronics, glass, tyres, interior)	2,944
Assembly and stamping labour (paid by the hour on the line)	560
Factory building depreciation, insurance and plant-management salaries	640
Delivery, showrooms and marketing for cars sold this quarter	480
Battery-research spending charged to the line this quarter	220
Retooling of the Model Y line, completed and paid last quarter	400
Memo: Model Y cars built this quarter	160,000 cars

Part B – Tuesday: the price war and the invisible split

Tuesday opens with a forwarded headline and one line from Reyes: "This is why I'm asking."

Tesla cuts U.S. prices of Models Y, X, S by \$2,000 (CNBC, 20 April 2024). By that cut, the base Model Y stood at **\$42,990** – roughly **\$23,000** below the cheapest Model Y's January 2023 sticker (then the \$65,990 Long Range), after a single \$13,000 cut that month and repeated smaller cuts since. Like-for-like, the Long Range itself was down about \$18,000, to \$47,990.

The question every analyst asked is the one on your desk: **if Tesla sells more cars at a lower price, does profit go up or down?** You cannot answer it without knowing how much of Tesla's cost is fixed and how much is variable – volume spreads fixed costs but adds variable ones. And Tesla, like every company, never publishes "our fixed cost is \$X". All the outside world sees is total cost and volume, period by period. Recovering the split from those totals is **cost estimation**.

Exhibit B-1 gives you the raw material: eight quarters of real Compustat data, 2023–24 – revenue, COGS, production and deliveries. Before any method, one judgement call: the file has both *production* (cars built) and *deliveries* (cars sold). Which is the right activity driver for COGS? Think about what COGS is before you look at the data; then note that the data agrees with you (Exhibit B-2's footnote).

The quick-and-dirty method is **high-low**: take the quarter with the highest activity and the one with the lowest, and assume the entire cost difference between them is variable:

$$b \text{ (variable cost per unit)} = (\text{Cost}_{\text{high}} - \text{Cost}_{\text{low}}) \div (\text{Units}_{\text{high}} - \text{Units}_{\text{low}}), \text{ then } a \text{ (fixed cost)} \\ = \text{Cost}_{\text{high}} - b \times \text{Units}_{\text{high}}.$$

It takes two minutes with a calculator – and it listens to only two quarters, throwing the rest away. If either extreme was an unusual quarter, the whole line is wrong, and high-low cannot even tell you how much to trust it.

Which is why the finance team had already fitted the line another way before you arrived. **Ordinary least squares (OLS) regression** draws the single straight line that sits as close as possible to *all* the points at once, and its two pieces are exactly what you need: the **intercept** is the fixed cost, the **slope** the variable cost per car. Their output is Exhibit B-2: an estimated fixed cost of **\$4,295m per quarter**, a variable cost of **\$31,951 per car**, an R^2 of **0.842** and a p-value of **0.0013**. In plain words: the line explains about 84% of the movement in quarterly COGS, and a relationship this tight is very unlikely to be luck – although with eight data points, treat any estimate as indicative rather than gospel. One caveat travels with it: the COGS column is Compustat's, and Compustat reports COGS with depreciation, amortisation and impairment shown separately rather than inside the line (\$5,368m in FY2024) – so the intercept *understates* Tesla's total fixed production cost.

Now the split earns its keep. **Cost–volume–profit** logic runs on one defined term: **contribution margin** – selling price minus variable cost per unit. It is *not* profit; it is what each car contributes towards a pile of fixed cost, and only once the pile is gone does contribution become profit. Break-even is simply the number of contributions it takes to fill the pile: $Q^* = \text{Fixed cost} \div \text{CM}$. For a revenue figure per car, the best the outside world offers is **\$54,599** – 2024 revenue of \$97,690m spread over 1,789,226 deliveries. Handle it with the label on: it folds energy, services and regulatory-credit revenue over the cars, so it sits well above the ~\$43k Model Y sticker. It is a proxy, not an average selling price.

Run the numbers (Question 4) and Tuesday's resolution emerges: against its COGS break-even, Tesla's actual 2024 delivery run-rate carries a **wide margin of safety** – deliveries could fall by more than half before quarterly contribution stopped covering the estimated fixed pile; Question 4 puts a number on it. On production economics, the price cut does not sink the company – *provided the volume holds*. The caveat belongs in the same sentence: this break-even covers the cost of *making* the cars only. SG&A and R&D – some \$9.7bn a year in Exhibit A-1 – still sit on top before real operating profit. Cost estimation has not given you certainty; it has given you a number to argue over, which is what the whole price-war debate was missing.

Assignment questions – Part B

1. Exhibit B-1 has both production and deliveries. Which is the right activity driver for COGS, and why?
2. High-low on Exhibit B-1 (work in \$m; deliveries are in cars): identify the high and low quarters by deliveries, then compute b and a . Which six data points did you just throw away?
3. Compare your high-low line with Exhibit B-2's OLS line (\$4,295m + \$31,951 per car). Why do they differ? Make the case for the line you would take to the CFO, and say what $R^2 = 0.842$ and $p = 0.0013$ each tell you, in plain words.
4. CVP with the OLS estimates: compute the contribution per car, the break-even volume per quarter, and the margin of safety against actual average 2024 deliveries. So: does the price cut, by itself, sink Tesla? State the one caveat about what this break-even does and does not cover.
5. ► Sensitivity: suppose revenue per car delivered falls a further 10%, to \$49,139. Recompute the contribution per car and the break-even. Then one paragraph: why might the straight line itself mislead outside the range of volumes you observed – and what does the “sticky costs” evidence (Anderson, Banker & Janakiraman 2003, *Journal of Accounting Research*) say about the symmetric $a + bX$ assumption?
6. A colleague ranks four possible drivers for quarterly COGS by how well each fits the eight quarters (Exhibit B-3) and recommends total revenue – “the highest R-squared on the list, 0.963”. Which driver

should you use to estimate Tesla's COGS cost function? Give, in one line each, the reason you reject the other three. For the revenue line, your reason must say what its fitted intercept of minus \$573m and its slope of 0.791 actually mean.

7. ► Tesla issues delivery guidance of 500,000 cars for next quarter. Using the estimated cost function in Exhibit B-2, forecast next quarter's COGS, then compute the projected quarterly contribution at the labelled revenue-per-car proxy. Exhibit B-4 lists six planning figures; three of them do not belong in this forecast – identify each and say why in one line. Finally, state the assumption your forecast relies on about the 500,000 figure, and name what still sits on top of contribution before Tesla would reach operating profit.

Exhibits – Part B

Exhibit B-1 – Eight quarters of real data, 2023–24 REAL – Compustat, quarterly, 2023–24. Note: Compustat reports COGS with depreciation, amortisation and impairment shown separately (\$5,368m in FY2024); a fixed-cost estimate built on this series therefore understates total fixed production cost.

Quarter	Revenue (\$m)	COGS (\$m)	Production (cars)	Deliveries (cars)
Q1 2023	23,329	17,772	440,808	422,875
Q2 2023	24,927	19,240	479,700	466,140
Q3 2023	23,350	17,937	430,488	435,059
Q4 2023	25,167	19,497	494,989	484,507
Q1 2024	21,301	16,359	433,371	386,810
Q2 2024	25,500	19,644	410,831	443,956
Q3 2024	25,182	18,837	469,796	462,890
Q4 2024	25,707	20,032	459,445	495,570

Exhibit B-2 – The finance team's regression output ESTIMATE computed from the REAL Compustat data in Exhibit B-1 (OLS, quarterly COGS in \$m against deliveries, 8 quarters). An estimate, not a figure Tesla discloses.

Output	Value
Intercept (fixed cost)	\$4,295m per quarter
Slope (variable cost)	\$31,951 per car
R ²	0.842
p-value (slope)	0.0013

Footnote: regressed on production instead of deliveries, R² collapses to 0.147.

CVP input: 2024 revenue per car delivered = \$97,690m ÷ 1,789,226 = **\$54,599** – revenue per car delivered includes energy, services and regulatory credits, so it sits above the ~\$43k Model Y sticker; a proxy, not an ASP.

Exhibit B-3 – A colleague's shortlist of drivers for quarterly COGS ESTIMATE – R², fitted intercept and slope computed from the REAL Exhibit B-1 data (8 quarters, OLS, quarterly COGS in \$m); the Supercharger row is SIMULATED (illustrative).

Proposed driver	R ²	Fitted intercept	Fitted slope	Note
Deliveries (cars sold)	0.842	\$4,295m	\$31,951 / car	the Exhibit B-2 line
Production (cars built)	0.147	\$11,153m	\$16,603 / car	–
Total revenue (\$m)	0.963	–\$573m	0.791 per \$ of revenue	highest R ² on the list

Proposed driver	R ²	Fitted intercept	Fitted slope	Note
Supercharger stalls energised	—	—	—	series recorded only from Q3 2023 (6 of the 8 quarters)

Exhibit B-4 — Next-quarter planning sheet (for the COGS forecast) SIMULATED — illustrative planning figures; each \$ magnitude is set near Tesla's real FY2024 quarterly run-rate (10-K SG&A \$5,150m/yr; regulatory-credit revenue ~\$2.76bn/yr) and labelled — not Tesla's actual plan.

Item	Figure
Delivery guidance, next quarter	500,000 cars
Estimated cost function (from Exhibit B-2)	COGS (\$m) = 4,295 + 0.031951 × cars
Revenue per car delivered (proxy, from Part B)	\$54,599
Allocated corporate SG&A, next quarter	\$1,290m
Battery-line retooling, paid last quarter	\$900m
Regulatory-credit revenue, next quarter	\$650m

Part C — Wednesday: one factory, five models, one pot of overhead

Wednesday morning, the question sharpens from “a Tesla” to “which Tesla”. The same factory floor you walked on Monday builds several models side by side — and someone in the pricing team has mailed around the per-car arithmetic with the subject line “why do we bother with the cheap ones?”

Stylised liberty, flagged: for a clean single-factory example, this part places all five models — Y, 3, S, X and Cybertruck — at Fremont. In reality the Cybertruck is built at Giga Texas, and the volumes here (1.23m cars a year) are stylised towards Tesla's global model mix — about double the real plant's ~650k installed annual capacity (per the 10-K). All internal figures in this part are simulated (Exhibits C-1 to C-3), and the margin levels are stylised at roughly 2.4× Tesla's real 17.9% FY2024 gross margin — read the distribution across models, not the level.

The pricing team's point looks unanswerable. Before any shared costs, a Model X clears \$67,000 per car over its direct materials (\$99,000 – \$32,000); a Model Y clears \$26,000 (\$44,000 – \$18,000). Pour the factory into premium cars, surely.

Except that between “price minus materials” and “profit” sits a **\$9,200m pot of shared overhead** — assembly lines, the battery plant, test labs, logistics, the building itself (Exhibit C-2). None of it can be traced to a single car; all of it must be *allocated* by a rule. And here is the uncomfortable discovery of Day 3: **whichever splitting rule you pick decides which model looks profitable.**

The simplest rule — and most firms' default — is the **traditional volume allocation**: one plant-wide rate per car, on the logic “more cars, more cost”. Divide \$9,200m by 1,230,000 cars and every model, from a Model 3 to a Cybertruck, carries the same **\$7,480** of overhead. On that rule the premium cars look spectacular: margins around 60%, against roughly 42% for the volume cars (Exhibit C-3, top rows).

But walk the floor before you believe the spreadsheet. A Cybertruck spends 25 hours in assembly against a Model 3's 10 — two and a half times as long. It carries a 123 kWh battery against 60. It goes through six test cycles against three. Does it *sip the same overhead* as a Model 3? Charging every car the same \$7,480 quietly makes the simple, high-volume cars pay for the complex, low-volume ones. That has a name — **cross-subsidisation** — and a cure.

Activity-based costing (ABC) splits the overhead into pools and charges each pool by the **driver** that actually causes it: assembly by assembly hours, the battery plant by kWh, testing by test cycles (Exhibit C-2

gives each pool and its driver total). Divide each pool by its driver total and you get a rate; charge each model for the drivers it consumes. The costing team ran exactly this before being pulled onto another project, leaving you Exhibit C-3 – complete except for the Cybertruck column, which is yours (Questions 2 and 3). The logic behind the drivers is Cooper and Kaplan's **cost hierarchy**: some overhead is caused by each *unit* (assembly hours, kWh), some by each *batch*, some by a *product line* existing at all, and some by the *facility* existing – and a volume rate wrongly treats all four levels as unit-level, which is precisely why complex low-volume products come out under-costed.

When the column is filled in, the reveal is not that anything makes a loss – nothing does – but where the money actually lives. Per car, the premium models still earn the most. In *total*, they are a garnish: **Model Y and Model 3 together carry about 86% of the annual profit** in this stylised Fremont world (directionally right for the real Tesla, where the 3/Y pair is ~95% of volume). “Most profitable per car” and “where the money is” are different questions, and ABC sharpens the difference – deflating the premium margins and exposing the Cybertruck as earning far less than its sticker suggests.

Late on Wednesday, Reyes reads the table and asks the natural next question: “*If the Cybertruck’s margin is the weakest, why not drop it?*” Treat that question with respect – and suspicion. ABC allocates the Cybertruck roughly \$485m of overhead a year, but **allocated is not avoidable**: most of the \$9,200m is shared, committed capacity – the building, the lines, the battery plant – that stays whether or not the Cybertruck exists. Kill the model and, in the worst case (none of its overhead avoidable), the full \$485m reloads onto the survivors at about +\$407 per car – at which point another model “looks weak”, and the road ends in the **death spiral**: dropping products one by one because a spreadsheet rule keeps re-spreading the same fixed pot. Which costs would *actually* disappear is a different question from anything you have computed today. It is Thursday’s question.

Assignment questions – Part C

1. Compute the plant-wide blended rate from Exhibits C-1 and C-2. Under it, show the profit per car for the Model 3 and the Model X (price – materials – overhead). On these numbers, should Tesla pour the factory into premium cars? Hold your answer until Question 4.
2. Compute two ABC rates by calculator, exactly as the exam would ask: the **battery systems** rate per kWh and the **assembly** rate per hour. (The other three rates are given in Exhibit C-3.)
3. Fill in the Cybertruck column of Exhibit C-3: its ABC overhead per car, profit per car and margin. Then, in one paragraph, use **cross-subsidisation** and the **cost hierarchy** (unit/batch/product/facility) to explain why the volume rate flattered it.
4. Using C-3’s annual columns: where does the profit actually come from, and how do you reconcile “a Model X earns \$56,749 a car” with “Model Y and Model 3 are ~86% of the total”? Then the trap: the CFO suggests dropping the Cybertruck on its weak ABC margin. Its allocated overhead is ~\$485m a year – explain, pool by pool, why allocated ≠ avoidable, what the worst case does to the surviving models, and where that road ends.
5. ► **Discovery.** (*Hypothetical single-plant quarter, labelled illustrative: price \$45,000, variable cost \$30,000 per car, fixed production overhead \$1,200m, sales fixed by demand at 120,000 cars.*) Your bonus rides on reported (absorption-costing) profit, and you are about to miss. Show that producing 150,000 cars instead of 120,000 lifts reported profit from \$600m to \$840m while variable-costing profit stays at \$600m; trace exactly where the missing \$240m went. Name the practice, explain why it is legal yet value-destroying, and connect it to Roychowdhury (2006).
6. Before this morning’s allocation, the \$9,200m overhead pot had to be assembled from a longer working list. Exhibit C-4 shows the eight lines a junior analyst started from; the pot actually used in Questions 1 to 3 was \$9,200m. Three of these lines do not belong in a Fremont product-overhead pot. Identify the three, give a one-line reason for each, and confirm that the remaining lines reconcile to \$9,200m. Does removing them change the plant-wide rate of \$7,480 you found in Question 1?
7. The CFO drops the Cybertruck anyway, on its weakest ABC margin. Take the worst case from Question 4 – that none of its overhead is avoidable – so its full ~\$485m reloads flat across the 1,190,000 surviving cars, at ~+\$407 per car. Add that \$407 to each surviving model’s ABC overhead from Exhibit C-3, recompute the four margins, and identify which model the same “drop the weakest margin” rule now points at. State the one assumption your answer rests on, and explain in two sentences why running the rule a second time is self-destructive.

Exhibits – Part C

Exhibit C-1 – Fremont product data (stylised) SIMULATED – volumes, prices, materials and consumption figures built from industry benchmarks and teardown reports. Stylised liberty flagged: all five models are placed at Fremont for a clean single-factory example; in reality the Cybertruck is built at Giga Texas, and the 1.23m total is roughly double the real plant's ~650k installed annual capacity (it leans towards Tesla's global model mix).

Model	Annual volume	Price	Direct materials	Assembly hrs/car	Battery kWh	Test cycles
Model Y	650,000	\$44,000	\$18,000	12	75	3
Model 3	500,000	\$39,000	\$15,000	10	60	3
Model S	20,000	\$89,000	\$28,000	18	100	5
Model X	20,000	\$99,000	\$32,000	20	100	5
Cybertruck	40,000	\$80,000	\$35,000	25	123	6
Total	1,230,000					

Exhibit C-2 – Overhead cost pools and driver totals SIMULATED – pool sizes based on auto-industry benchmark cost shares. Note: logistics and facilities are charged per vehicle as proxies – logistics is really a batch-level cost (cars ship in multi-car transporter loads) and facilities a facility-level cost spread by volume, not a true square-foot driver.

Cost pool	Annual cost (\$m)	Cost driver	Driver total
Assembly	2,400	Assembly hours	14.56m hours
Battery systems	3,200	Battery kWh	87.67m kWh
Quality testing	800	Test cycles	3.89m cycles
Logistics	1,200	Vehicles	1.23m vehicles
Facilities	1,600	Vehicles	1.23m vehicles
Total	9,200		

Exhibit C-3 – The costing team's part-finished comparison SIMULATED – same stylised levels as C-1/C-2. Note carried from the costing team: margins here run ~2.4× Tesla's real 17.9% FY2024 gross margin – read the distribution across models, not the level. Three ABC rates are given: quality testing \$205.66 per cycle, logistics \$975.61 per vehicle, facilities \$1,300.81 per vehicle; the assembly and battery rates are Question 2, and the Cybertruck column is Question 3.

	Model Y	Model 3	Model S	Model X	Cybertruck
Traditional overhead per car	\$7,480	\$7,480	\$7,480	\$7,480	\$7,480
ABC overhead per car	\$7,609	\$6,732	\$9,922	\$10,251	[?]
Margin – traditional	42%	42%	60%	60%	47%
Margin – ABC	42%	44%	57%	57%	[?]
Annual ABC profit (\$bn)	12.0	8.6	1.0	1.1	[?]
Share of total ABC profit	50%	36%	4%	5%	[?]

Total annual ABC profit across all five models $\approx$ \$24.1bn. “Margin” = (price – materials – overhead) $\div$ price.

Exhibit C-4 – The analyst’s working list (before the pot was fixed) SIMULATED – illustrative Fremont-scale figures, not Tesla actuals.

Candidate line	\$m
Assembly	2,400
Battery systems	3,200
Quality testing	800
Logistics	1,200
Facilities	1,600
Head-office administration and national advertising	1,900
Paint-shop retooling – a one-off project completed last year	600
Battery-plant electricity (metered separately; forms part of the Battery systems pool above)	300
Working-list total	12,000

Part D – Thursday: make or buy – the battery question

Thursday’s diary says “board pre-read”, and the pre-read contains a memo from a colleague that is going to teach you the most important single idea of the week – by being wrong.

The board question: should Tesla make battery cells in-house, or buy them from Panasonic? The memo’s answer:

“Our in-house full cost is \$70 variable plus \$800m of annual fixed spread over 87.7m kWh – call it \$79/kWh. Panasonic lands at \$90. Make: we save \$11.”

It sounds fine. The arithmetic is even correct. And the reasoning can be exactly backwards, because the memo never asked *which costs are allowed into the comparison*.

The rule that sorts it out has two gates. A cost is **relevant** to a decision only if it is (1) **future** – not yet incurred, still changeable – and (2) **different between the alternatives**. Fail either gate and the cost is noise for *this* decision, however large and however painful.

Gate 1 dispatches the biggest number on the table. Tesla reported **\$6.2bn invested in Gigafactory Nevada by early 2023, with a \$3.6bn expansion underway** (real figures, labelled in Exhibit D-1). All of it is **sunk** – gone whichever way the cell decision goes – so none of it may touch the choice. What *can* matter about the factory is only what is still changeable: its resale value, or what else the floor could earn.

Gate 2 dispatches the memo’s \$800m. For the decision actually on the table – *where should the next batch of cells come from, on the existing line* – the \$800m of ongoing fixed cost is there whether the next batch is made or bought. It does not differ, so it does not count. The relevant comparison is **\$70 versus \$90 per kWh: making saves \$20**. The memo’s conclusion (make) happens to survive; its reasoning (full cost) does not – and Question 3 shows how quickly that luck runs out, because the moment extra demand needs a **new** line at \$400m a year, that fixed cost *is* future-and-different, and making only wins beyond a break-even volume of extra demand.

Midday brings an interruption that runs on the same rule. A rental company offers to buy **50,000 Model 3s at \$30,000 each** – \$9,000 below list (Exhibit D-2, illustrative). Full cost per car is \$33,000, so the offer “loses \$3,000 a car”... on a full cost that includes \$8,000 of allocated fixed overhead that exists either way. With spare capacity, the relevant comparison is price against *variable* cost, and the order adds a quarter of a billion dollars of contribution. At *full* capacity the logic flips: each special car displaces a full-price sale, and the displaced contribution – invisible on any invoice – is a real **opportunity cost**. Sunk costs are visible but irrelevant; opportunity costs are invisible but relevant. Instinct weights them exactly backwards.

Then, late afternoon, the board widens the question – and the same \$800m changes sides. Not “where does the next batch come from?” but “**should Tesla be in the cell business at all, or exit entirely and clear the floor?**” Against *that* alternative, the \$800m of ongoing cash fixed cost is avoidable – closing the line stops it – so it now passes the “differs” gate. And the cleared floor could earn roughly **\$1,100m a year of Megapack contribution** (profit forgone, not revenue – Exhibit D-1, illustrative), which is an opportunity cost of staying. Same dollars, different question, different relevance – and, as Question 5 shows, a different answer.

The week’s quietest lesson lands here: the numbers narrow the choice; judgement finishes it. The real Tesla both makes cells (its 4680 programme) *and* buys them (Panasonic, LG, CATL) – because supply security and proprietary battery technology are worth real money that appears in no relevant-cost table.

Assignment questions – Part D

1. State the relevant-cost principle (future **and** differs). Apply gate 1 to the Gigafactory’s ~\$6.2bn invested to date: is it relevant to make-versus-buy? What, if anything, about the factory can still be relevant?
2. Your colleague’s memo argues: “full cost in-house = \$70 + \$800m ÷ 87.7m kWh ≈ \$79/kWh, versus \$90 from Panasonic – make.” For the decision actually on the table – where the next batch of cells comes from, on the existing line – rebuild the comparison with relevant costs only, and explain which gate the \$800m fails and why the conclusion happens to survive while the reasoning does not.
3. Extra demand beyond current capacity would need a new \$400m-a-year line. Find the break-even extra volume and state the decision rule either side of it.
4. The special order (Exhibit D-2): show that the full-cost view rejects it and the relevant view accepts it with spare capacity. Then the hinge: at full capacity, what does each special car displace? Give the decision and the minimum acceptable price in each capacity state.
5. ► The board widens the question: exit the battery business entirely and clear the floor. Show that the \$800m now passes the “differs” test and that the forgone Megapack contribution is a relevant opportunity cost, and compute the relevant totals of MAKE versus BUY. Two sentences on why the same \$800m changed status – and two qualitative factors that explain why the real Tesla does both.
6. Your colleague, challenged on the \$79 memo, produces the fuller worksheet in Exhibit D-1 (further lines). For the decision in Question 2 – where the next batch of cells comes from, on the existing line – go through every figure now on the worksheet and sort it into “counts” or “ignore for this decision”. For each figure you ignore, name which test it fails: not future, or does not differ. State the one assumption about the existing line that lets you set the \$800m and the depreciation aside. Then work out what the padded “full cost” per kWh would have told you, and explain in two sentences why the relevant comparison disagrees.
7. ► Question 4 gave you two clean cases: all spare capacity, or none. Reality is usually in between. Take Exhibit D-2 with the Question 7 note – the Model 3 line can build 500,000 cars a year and currently sells 470,000, and the rental company still wants 50,000 cars at \$30,000. Work out how many of the 50,000 use idle capacity and how many displace full-price sales, and hence the relevant profit or loss of accepting the whole order. Give the minimum price at which you would take the whole order. Then find how much spare capacity the line would need for the order to be worth taking at \$30,000, and state the assumption your answer depends on.

Exhibits – Part D

Exhibit D-1 – Battery cell cost data (make versus buy) SIMULATED – benchmarked on the BloombergNEF battery cost index and industry reports. **Scope flag: these are cell-level variable costs – do not compare them to ~\$115–140/kWh pack-level prices (BloombergNEF 2024), which add pack assembly, housing and battery management; a different scope.**

Item	\$/kWh
In-house cell, variable cost	70
– raw materials	45
– direct labour	8

Item	\$/kWh
– manufacturing overhead (variable)	12
– quality control	3
– logistics	2
Panasonic, landed	90
– quoted price	85
– logistics	5

Other data	Figure
In-house ongoing cash fixed cost (utilities, maintenance, salaried staff, leases) – avoidable only if the line closes; excludes the sunk build	\$800m/yr
Annual cell demand (the 87.67m kWh battery driver total from Exhibit C-2; 87.7m in round figures)	87.7m kWh

Further lines from the colleague's worksheet (SIMULATED – illustrative; the fuller worksheet the colleague produces when the \$79 memo is challenged. Not all of these belong in the make-or-buy comparison.)

Item	\$/kWh
Corporate head-office cost allocated to the cell line (spread by kWh)	6
Depreciation of the existing cell line (accounting spread of the sunk Gigafactory build)	9
Panasonic's list quote before Tesla's negotiated volume discount	95

*Context (REAL): Tesla reported \$6.2bn invested in Gigafactory Nevada by early 2023, with a \$3.6bn expansion underway. Illustrative additions (SIMULATED): a new production line would cost \$400m/yr in fixed costs; the freed floor could earn ~\$1,100m/yr of Megapack **contribution** (profit forgone, not revenue).*

Exhibit D-2 – The rental-fleet special order SIMULATED – labelled illustrative; simplified Model 3 costs.

Item	Figure
Offer	50,000 Model 3s at \$30,000 per car
List price	\$39,000
Variable cost per car	\$25,000
Allocated fixed overhead per car	\$8,000
"Full cost" per car	\$33,000

For Question 7 (SIMULATED – illustrative): the Model 3 line is built for its planned 500,000 cars a year (Exhibit C-1) and is currently running at 470,000. The rental order would be additional to current sales.

Part E – Friday: the miss – was it the market or the factory?

Friday, 8:05 a.m. The quarter's flash numbers land, and they land badly: operating profit is **\$352m below plan**. Reyes's message beats the coffee to your desk:

"Before this afternoon: was that the market, or the factory? Three bullets. Don't bring me a spreadsheet — bring me who answers for what."

*(The figures in this part are simulated — standards constructed with Tesla's real Q4 2024 as the base, with one stated simplification: cars made = cars sold, so a single volume drives both revenue and cost. The \$9,750m fixed pot is **all** of Tesla's fixed costs — fixed production overhead plus R&D plus SG&A — so the bottom line approximates quarterly operating profit; the magnitude survives a reality check, since Tesla's reported Q4 2024 operating income was \$1,583m (~\$1.6bn) and the case's lands nearby. Reminder: this part is examined at the **final** exam, not the mid-session.)*

Exhibit E-1 lays plan against reality: 495,570 cars delivered against a 500,000 budget; revenue per car a touch soft; materials \$500 a car over standard; labour \$300 under; overheads a whisker adrift. The temptation is to compare actual profit straight against the original budget and start assigning blame. Resist it. The original — **static** — budget was built for 500,000 cars, and the company sold 495,570; comparing across different volumes punishes the factory for the sales team's shortfall. So the first move of any variance analysis: **flex the budget** — restate the plan at the volume actually achieved — and only then compare.

Flexing splits the \$352m miss into two honest pieces. The **sales-volume variance** — the profit lost purely because 4,430 fewer cars were sold, each carrying its budgeted contribution, which is why this variance is **measured on a contribution basis** — and the **flexible-budget variance**, everything that went differently *per car* at the volume actually sold. The first belongs to the demand side of the business; the second to prices and efficiency.

The flexible-budget variance then decomposes line by line (Question 2) — and the decomposition rewrites the story the headline told: the miss is *not* simply "the factory ran badly". Question 2 quantifies which lines actually carry it — and which line, surprisingly, pushes the other way.

But even the materials line is two stories wearing one number, and the tool that separates them is the **standard cost card** (Exhibit E-2). The card sets, in advance, a standard price and a standard quantity per car for each input. Compare actuals against it and the materials line splits into a **price variance** — the factory paid \$50 a unit instead of \$45 — and a **quantity variance** — it used 370 units a car instead of 400. Those two pieces have different owners: purchasing (and the world lithium price) owns the first; the production line owns the second. Net them and you would congratulate or blame the wrong people: the netted "–\$500 a car" was hiding a large price hit almost three-quarters offset by a genuine efficiency gain worth celebrating.

What goes to Reyes at 3 p.m. is therefore not a data dump but a judgement, built on two principles. **Controllability**: hold managers to what they control — blaming the plant for lithium prices or a demand shortfall is unfair, demotivating, and teaches people to pad next year's budget. **Management by exception**: spend attention only on variances that are *big and controllable* — chase the materials story and the volume question (was it demand, or could we not build enough?), reinforce the labour win, and let the small wobbles wait.

And one last discipline before the memo goes out, supplied by Question 5: a variance is only as trustworthy as the budget behind it, and any control system that watches only profit invites the very trick you discovered on Wednesday.

Assignment questions — Part E

1. From Exhibit E-1, compute the static-budget profit, the flexible-budget profit at actual volume, and actual profit (work in \$m). Split the total variance into the **sales-volume variance, measured on a contribution basis**, and the **flexible-budget variance**. Why must you flex before blaming anyone?
2. Decompose the –\$248m flexible-budget variance line by line at the actual volume of 495,570 cars. Which two numbers would you lead the briefing with?
3. Using Exhibit E-2, split the materials line into its **price** and **quantity** variances, per car and in \$m. Why do the two pieces have different owners, and what management conversation was the netted line hiding?
4. Split the labour line into **rate** and **efficiency** variances — then write the CFO's three-bullet memo: what drove the miss (numbers, F/U), which variances deserve management attention and who owns each (controllability, management by exception), and one recommendation for next quarter.

5. ► Two challenges before the memo goes out. **(a) Budget quality:** Musk's public targets are famously aspirational – the "20 million cars by 2030" goal was quietly dropped. What does a variance against a soft or moon-shot budget actually measure, and what does the target-ratchet evidence (Bouwens & Kroos 2011, *Journal of Accounting and Economics*) predict managers will do near year-end? **(b) Gaming:** recalling Part C's discovery question (Question 5), explain how the plant could have "fixed" this quarter's miss without selling a single extra car under absorption costing – and why a well-designed control system must watch production alongside profit.
6. Exhibit E-3 lists four further figures that reached your desk after the flash numbers. **(a)** For each figure, say in one line whether it belongs in this quarter's operating-profit variance, and if it does not, why not. **(b)** Now take the fixed-cost figure – the one that does carry information. Separate the recurring cost base from the one-off charge, state the assumption your treatment depends on, and say how the split changes the management-by-exception verdict on the –\$50m fixed variance.
7. ► Exhibit E-4 opens up the variable-overhead line the briefing waved through as a small wobble. **(a)** Which of the two hour totals is the right activity driver for variable overhead – and why is the other one on the sheet? **(b)** Derive the actual rate per machine-hour. **(c)** Split the variable-overhead line into its spending (rate) and efficiency variances – per car and in \$m – and show they reconcile to the –\$49.6m line. **(d)** In one sentence, what does this line do to the rule "ignore the small variances"?

Exhibits – Part E

Exhibit E-1 – Q4 budget versus actual SIMULATED – standards constructed with Tesla's real Q4 2024 as the base. Simplifications stated: cars made = cars sold; the fixed pot is ALL fixed costs (fixed production overhead + R&D + SG&A), so the bottom line ≈ quarterly operating profit – a magnitude that survives a reality check against Tesla's reported Q4 2024 operating income of \$1,583m.

Item	Budget	Actual
Deliveries (cars)	500,000	495,570
Revenue per car delivered*	\$52,000	\$51,900
Direct materials per car	\$18,000	\$18,500
Direct labour per car	\$4,500	\$4,200
Variable overhead per car	\$6,000	\$6,100
Fixed costs (\$m)	9,750	9,800
Derived: variable cost per car	\$28,500	\$28,800
Derived: contribution per car (budget)	\$23,500	–

*Proxy label carried from Part B: total revenue spread over cars delivered, including energy, services and regulatory credits – not an ASP.

Exhibit E-2 – The standard cost card SIMULATED – labelled, like the rest of this part's figures.

Input	Standard	= per car	Actual this quarter
Direct materials	400 units @ \$45	\$18,000	370 units @ \$50
Direct labour	60 hours @ \$75	\$4,500	50 hours @ \$84

Exhibit E-3 – Four further figures on the analyst's desk Mixed sources – each row labelled REAL or SIMULATED. These figures reached your desk after the flash numbers; decide which, if any, belong in the quarter's operating-profit variance. The \$9,800m actual fixed cost in Exhibit E-1 is unchanged – row 4 is a component of it, not an addition to it.

#	Item	Figure	Label
1	Tesla operating profit, prior quarter (Q3 2024)	~\$2,700m	REAL – Tesla Q3 2024

#	Item	Figure	Label
2	Fremont paint-shop line (installed 2023), remaining book value	~\$600m	SIMULATED – illustrative
3	Regulatory-credit revenue booked in Q4	~\$690m	REAL – Tesla Q4 2024
4	One-off restructuring/severance charge sitting <i>inside</i> the \$9,800m actual fixed costs (not in the budget)	\$80m	SIMULATED – illustrative

Exhibit E-4 – Variable-overhead standard card and Q4 actuals *SIMULATED – an illustrative internal driver split, consistent with the rest of this part's figures; variable overhead per car (\$6,000 budget, \$6,100 actual) is carried unchanged from Exhibit E-1. The actual rate per machine-hour is Question 7.*

Item	Standard	Actual
Variable-overhead rate per machine-hour	\$1,000	[?]
Machine-hours per car	6.0	5.0
<i>Variable overhead per car</i>	<i>\$6,000</i>	<i>\$6,100</i>

Quarter totals (actual)	Figure
Total variable overhead	\$3,023m
Total machine-hours	2.478m
Total direct-labour hours	24.78m

Friday, 4 p.m.

The briefing slide you finally show Reyes has one sentence at the top: *"It depends which cost you mean – and here is which one you should mean, for each decision on your desk."*

Underneath it, the week:

- **Monday:** the published accounts say Tesla as a whole made money – \$17.45bn gross, \$7.08bn operating – but they contain no cost of a *Model Y*. Cost object first; then the classification that matches the question.
- **Tuesday:** for *"does the price war sink us?"*, the estimated fixed/variable split – a ~\$22,648 contribution per car against a ~\$4.3bn quarterly fixed pile – gives Tesla a margin of safety of roughly 58%: actual deliveries run about 2.4 times the COGS break-even volume. The cut survives if volume holds; SG&A and R&D still sit on top.
- **Wednesday:** for *"which model makes the money?"*, the allocated full cost – done properly, with ABC – shows the volume cars carrying ~86% of the profit while the premium badges supply the garnish.
- **Thursday:** for *"make or buy?"*, only the avoidable and opportunity costs count – and the same \$800m can be irrelevant to one question and decisive for the next.
- **Friday:** for *"who answers for the miss?"*, the standard cost – flexed to actual volume, split into price and quantity, judged by controllability.

Use the wrong cost for the day's decision and you hand a CFO a confidently wrong answer. Use the right one and the numbers do what management accounting is for: they narrow the choice – and then judgement finishes it.

Selected references woven through the case: Anderson, Banker & Janakiraman (2003), "Are selling, general, and administrative costs 'sticky'?", Journal of Accounting Research; Roychowdhury (2006), "Earnings management through real activities manipulation", Journal of Accounting and Economics; Bouwens & Kroos (2011), "Target ratcheting and effort reduction", Journal of Accounting and Economics.